

Matematisk Statistik

Sampling Distributions

Lars Nørvang Andersen

Institut for Matematik
Aarhus Universitet

Stikprøvefunktioner og sampling distributions - terminologi

Når vi har en statistisk model, hvor data $\mathbf{x} = (x_1, \dots, x_n)^*$ er udfald af en stokastisk vektor $\mathbf{X} = (X_1, \dots, X_n)^*$ kaldes en funktion af $T(\mathbf{X})$ (f.eks. estimatorer og testorer) generelt for en *stikprøvefunktion*. Fordelingen af $T = T(\mathbf{X})$ kaldes for *sampling distribution/stikprøvefordelingen* og spredningen af T kaldes *standard error*.

Eksempler:

Example 4.1 (næste slide)

Example 4.2 Fordeling af $\bar{X}$ når $X_1, \dots, X_n \stackrel{i.i.d}{\sim} E(\frac{1}{15})$

Example 4.3 og 4.4 Fordeling af $X_{(12)}$ (maksimum) når $X_1, \dots, X_{12}$ er $U[0, 1]$.

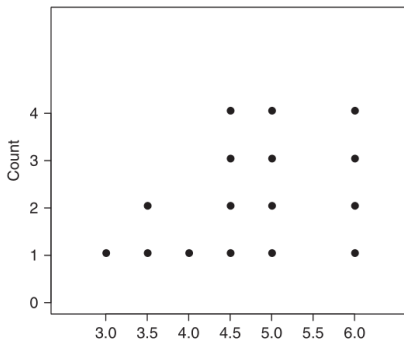
Example 4.1

Population: $w_1 = 3, w_2 = 4, w_3 = 6, w_4 = 6$.

$\mathbb{P}_e(X = 3) = \mathbb{P}_e(X = 4) = 1/4, \mathbb{P}_e(X = 6) = 1/2$

Sampling distribution of sample mean: $(X_1 + X_2)/2$.

Figure 4.3 Dot plot for sampling distribution of sample means.



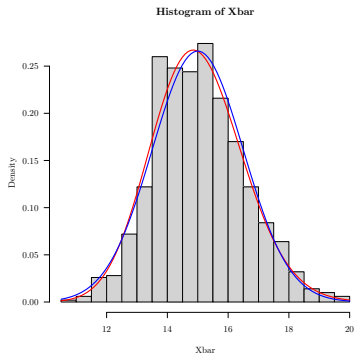
Sample mean	w_1, w_1	w_1, w_2	w_1, w_3	w_1, w_4	w_2, w_1	w_2, w_2	w_2, w_3	w_2, w_4
	3	3.5	4.5	4.5	3.5	4	5	5
Sample mean	w_3, w_1	w_3, w_2	w_3, w_3	w_3, w_4	w_4, w_1	w_4, w_2	w_4, w_3	w_4, w_4
	4.5	5	6	6	4.5	5	6	6

Example 4.2 - sampling distribution ved simulation

```
Xbar <- numeric(1000) # space for results (vector of 0's)
for (i in 1:1000) {
  x <- rexp(100, rate = 1/15) # draw random sample of size 100
  Xbar[i] <- mean(x) # compute mean, save in position i
}
hist(Xbar, freq=FALSE, breaks = 20)

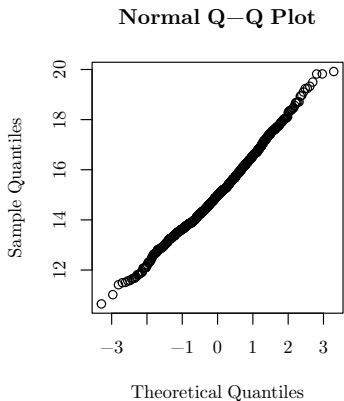
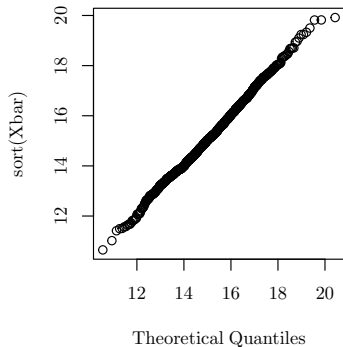
# Eksakt fordeling
curve(dgamma(x, shape=100, rate=100/15), add=TRUE, lwd=2, col="red")

# Normal approksimation
curve(dnorm(x, mean=15, sd=15/10), add=TRUE, lwd=2, col="blue")
```



Example 4.2 - sampling distribution ved simulation

```
plot(qgamma( (1:1000-0.5)/1000, shape=100, rate=100/15), sort(Xbar), xlab="Theoretical Quantiles")  
qqnorm(Xbar)
```



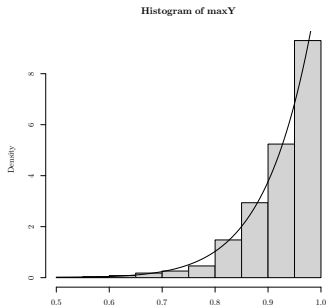
Example 4.4

$X_1, X_2, \dots, X_{12} \stackrel{\text{i.i.d.}}{\sim} U[0, 1]$

$$F_{X_{(12)}}(a) = 12a^{11}$$

```
maxY <- numeric(1000)
for (i in 1:1000)
{
  y <- runif(12) # draw random sample of size 12
  maxY[i] <- max(y) # find max, save in position i
}
hist(maxY, freq=F)

fun <- function(a) {12*a^11}
curve(fun, add=TRUE, lwd = 2)
```



Calculating Sampling Distributions - exact calculations

I visse tilfælde, kan vi benytte egenskaber for fordelingen af $\mathbf{X}$ til at udlede fordelingen af $T(\mathbf{X})$.

Corollary A.2 [‡] : $X_1, \dots, X_n \stackrel{i.i.d}{\sim} N(\mu, \sigma^2) \Rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

Theorem B. 16 [§] : Stadig $X_1, \dots, X_n \stackrel{i.i.d}{\sim} N(\mu, \sigma^2)$

- 1) $\bar{X}$ og $s^2(\mathbf{X})$ er uafhængige stokastiske variable.
- 2) $(n-1)s^2(\mathbf{X})/\sigma^2 \sim \chi^2(n-1)$

Theorem 4.1 [¶] : For en i.i.d. følge af kontinuerte stokastiske variable $X_1, \dots, X_n$ gælder at

$$f_{X_{(1)}}(x) = n(1 - F(x))^{n-1}f(x) \quad \text{og} \\ f_{X_{(n)}}(x) = nF^{n-1}(x)f(x).$$

Dette giver fordeling af MLE for $X_1, \dots, X_n \stackrel{i.i.d}{\sim} U[0, \beta]$

Example 4.6: $X_1, \dots, X_n$ uafhængige med $X_i \sim \text{Pois}(\lambda_i)$ så er $X_1 + X_2 + \dots + X_n \sim \text{Pois}(\sum_{i=1}^n \lambda_i)$

[‡]Corollary A.5.3 i [Chihara and Hesterberg(2018)]

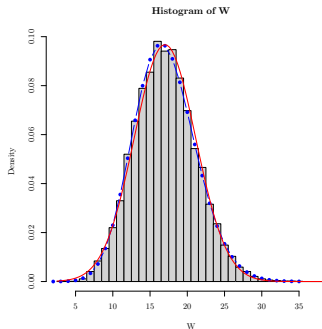
[§]Theorem B.10.5 i [Chihara and Hesterberg(2018)], Thm 8.3 i [Pishro-Nik(2014)]

[¶]Theorem 4.2.1 i [Chihara and Hesterberg(2018)]

Example 4.6

```
set.seed(1)
X <- rpois(10^4, 5)
Y <- rpois(10^4, 12)
W <- X + Y
hist(W, prob = TRUE, breaks = 2:38+0.5) # prob = TRUE, scales hist to 1
lines(2:35, dpois(2:35, 17), type = "b", col = "blue", pch = 16) # overlay pmf for Pois(17)

curve(dnorm(x, mean = 17, sd = sqrt(17)), lwd = 2, add = TRUE, col = "red")
```



The Central Limit Theorem

Central limit theorem (Den central grænseværdisætning) siger at for en stikprøve med middelværdi μ og varians σ^2 er $\bar{X}$ approksimativt normalfordelt:

Sætning (Thm. 4.2 (4.3.1 i [Chihara and Hesterberg(2018)]))

Let $X_1, X_2, \dots, X_n$ be independent, identically distributed random variables with mean μ and variance σ^2 , both finite. Then for any constant z ,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z \right) = \Phi(z)$$

where Φ is the cdf of the standard normal distribution.

Mange alternative opskrivinger - blandt andet

$$Z \stackrel{\text{def}}{=} \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mathbb{E}[\bar{X}]}{\text{SE}[\bar{X}]}$$

Anvendelser af CLT

Example 4.7

Lad $X_1, X_2, \dots, X_{30}$ være en stikprøve fra gamma fordelingen med form-parameter $r = 5$ og rate-paramter $\lambda = 2$. Benyt CLT til at approksimere $\mathbb{P}(\bar{X} > 3)$.

Vi finder

$$\mathbb{P}(\bar{X} > 3) \approx \mathbb{P}(Z > 2.4495) = 1 - \Phi(2.4495) = 0.0072.$$

Vi kan også benytte simulation the at finde en approksimation til $\text{Prob}(\bar{X} > 3)$.

Example 4.8

Er det plausibelt at $\bar{x} = 21.9$, hvis vi står med en stikprøve med $n = 17$ observationer fra Gamma(100, 5)-fordelingen?

Anvendelser af CLT - Normalapproximation for binomialfordelingen

Sætning (Corollary 4.2 (4.3.2 i [Chihara and Hesterberg(2018)]))

Let $X \sim \text{Binom}(n, p)$ være en binomialfordelt stokastisk variabel og lad $\hat{p} = X/n$ være frekvensen af "succes"-er. Hvis både np og $n(1 - p)$ er store, så er fordelingen af $\hat{p}$ approksimativt normal, med middelværdi p og spredning $\sqrt{p(1 - p)/n}$.

Tilsvarende er fordelingen af X approksimativt normal med middelværdi np og spredning $\sqrt{np(1 - p)}$.

Eksempler:

Example 4.10

Hvis 28% af >25-årige har en bachelorgrad, hvad er sandsynligheden for at mindst 30% af deltagerne i en stikprøve med $n = 64$ individer har en bachelorgrad?

Example 4.11

Lad $X \sim \text{Binom}(120, 0.3)$ beregn $\mathbb{P}(X \leq 25)$.

Bibliografi

 Laura M Chihara and Tim C Hesterberg.

Mathematical statistics with resampling and R (2nd edition).
John Wiley & Sons, 2018.

 H. Pishro-Nik.

Introduction to Probability, Statistics, and Random Processes.
Kappa Research, LLC, 2014.
ISBN 9780990637202.

URL

https://books.google.dk/books?id=3yq_oQEACAAJ.